

Personal Solutions to Problem Sheet 2 QFT by David Tong

Baset

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1.

According to the definitions made:

$$\begin{aligned}[a_n, a_m] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m + \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{i[q_n, p_m] + i[p_n, q_m]\} = 0\end{aligned}\tag{1}$$

$$\begin{aligned}[a_n, a_m] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m + \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{i[q_n, p_m] + i[p_n, q_m]\} = 0\end{aligned}\tag{2}$$

$$\begin{aligned}[a_n^\dagger, a_m^\dagger] &= [\sqrt{\frac{\omega_n}{2}}q_n - \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m - \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{-i[q_n, p_m] - i[p_n, q_m]\} = 0\end{aligned}\tag{3}$$

$$\begin{aligned}[a_n, a_m^\dagger] &= [\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n, \sqrt{\frac{\omega_m}{2}}q_m - \frac{i}{\sqrt{2\omega_m}}p_m] \\ &= \frac{1}{2}\{-i[q_n, p_m] + i[p_n, q_m]\} = \delta_{nm}\end{aligned}\tag{4}$$

Using 4:

$$\begin{aligned}H &= \sum_n \frac{1}{2}\omega_n(a_n a_n^\dagger + a_n^\dagger a_n) = \sum_n \omega_n(a_n^\dagger a_n + \frac{1}{2}) \\ &= \sum_n \omega_n((\sqrt{\frac{\omega_n}{2}}q_n - \frac{i}{\sqrt{2\omega_n}}p_n)((\sqrt{\frac{\omega_n}{2}}q_n + \frac{i}{\sqrt{2\omega_n}}p_n) + \frac{1}{2}))\end{aligned}\tag{5}$$